

Perspectives on Thermoelectricity in Layered and 2D Materials

Jing Wu, Yabin Chen, Junqiao Wu,* and Kedar Hippalgaonkar*

Layered materials have garnered immense interest due to their unique electronic, mechanical, thermal, and optoelectronic properties. In these materials, atomically thin layers are held together by van der Waals (vdW) interactions, allowing single layers to be isolated and studied as 2D materials. New theoretical insights as well as novel experimental techniques have empowered recent developments in both thermoelectric physics and performance of 2D materials. In this Review, some of the key ideas are surveyed and open questions are discussed, laying out a vision for original discoveries in this still-emerging field of applied physics. Here, graphene, transition metal dichalcogenides, black phosphorus, topological insulators as well as intercalated compounds are in focus. With these materials as a platform, how band structure, hydrodynamics, metal–insulator transitions, anisotropy, and surface states affect phonons and electrons and manifest in their thermoelectric properties is analyzed.

1. Introduction

Thermoelectrics is the field of study where a temperature gradient can be converted to electrical power and vice versa. The efficiency of a thermoelectric material is directly proportional to its Figure of merit (ZT), defined as $S^2\sigma T/\kappa$, where S is the Seebeck coefficient or thermopower, σ is the electrical conductivity, and κ is the thermal conductivity, including both from electrons (κ_e) and lattice vibrations (phonons, κ_l) as heat carriers at a thermodynamic temperature, T . However, the optimization of these three parameters (S , σ , κ) to achieve high ZT has become the key challenge due to their interdependence. They are all strongly dependent on the material's electronic structure and carrier concentration. For example, as the carrier concentration in a material is increased, the electrical conductivity increases also, but

the Seebeck coefficient drops as shown in Figure 1a.^[1,2] At the same time, the thermal conductivity from electrons (κ_e) will also increase according to the Wiedemann–Franz law. As a result of these interdependencies, good thermoelectric materials are typically narrow-bandgap semiconductors. This is because wide-bandgap semiconductors are relatively difficult to be highly doped,^[3] hence limiting their ability to achieve optimized electrical conductivity for high powerfactor ($S^2\sigma$). They are also typically composed of light elements, which translates to relatively high thermal conductivity. In the other extreme, metals or semi-metals have too large a carrier density or sometimes bipolar carriers, and their density of states (DOS) typically varies slowly at the Fermi level, which results in a low Seebeck coefficient. Addi-

tionally, to first order, the Seebeck coefficient typically peaks at $S_{\max} \sim E_g/2eT_{\max}$, where E_g is the bandgap and $T_{\max}$ is the temperature at which $S_{\max}$ is located, although more accurate estimates can be made.^[4] Therefore, narrow-bandgap semiconductors composed of heavy elements have been the focus in the search for high-performance thermoelectric materials, especially for low-to-intermediate temperature applications.

However, layered materials could offer a new platform where relatively high electrical conductivity coexists with high Seebeck coefficient. One of the advantages for layered materials is the quantum size effect.^[5] The fact that layered materials are atomically thin layers held together by weak vdW forces indicates that free carriers residing in each layer experience strong quantum confinement, akin to that in 2D. This is especially true for monolayers that can be isolated from the layered materials. As a result, the DOS of these 2D (or quasi-2D) systems significantly deviate from that of a 3D, bulk material, exhibiting strong anisotropy. For example, in the basal plane, sharp rises and peaks could develop in the DOS, different from the smooth DOS in 3D materials. Such rapidly varying DOS (with respect to electron energy E , i.e., high $d\text{DOS}/dE$) is beneficial for thermoelectricity. This is because when the Fermi level (E_F) is tuned to these positions in the DOS, a large asymmetry exists between hot ($E > E_F$) and cold ($E < E_F$) electrons, hence more effectively transporting entropy per charge, resulting in a higher Seebeck coefficient without a large sacrifice in the electrical conductivity. Meanwhile, the tunability of the Fermi level of the layered materials can be accessed much more easily than bulk materials by introducing intercalated ions or an electric field effect which assists in optimizing the Seebeck coefficient.^[6–10] Another

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potential benefit of layered materials is that this geometry makes it easier to separately engineer the transport of thermal phonons from that of free charge carriers. Also, the scattering times of electrons and phonons can be judiciously tuned by the change in dimensionality. This can be done in the classical picture by directly affecting the mean free path (average distance between collisions of quasiparticles) or in coherent and/or hydrodynamic ways by controlling feature sizes close to the phonon and/or electron wavelengths.

Due to the nanoscale nature of layered 2D materials, the thermoelectric measurement techniques are different from those employed for bulk materials. So, before starting the discussion about the thermoelectric properties, we would like to give a brief introduction to the experimental thermoelectric measurement techniques of nanoscale layered 2D materials. The main challenge to measuring the thermoelectric power-factor ($S^2\sigma$) is to obtain the Seebeck coefficient $S = -V_{\text{TEP}}/\Delta T$, which is defined as the ratio of open-circuit voltage V_{TEP} to the temperature difference ΔT along the 2D materials. The direct measurement of this Seebeck coefficient in 2D materials is possible by using locally fabricated microresistance thermometers across the 2D material as shown in Figure 1b. Such a technique has been used for many 2D materials including graphene^[8,10] as well as 1D nanowires^[11–13] and nanotubes.^[14]

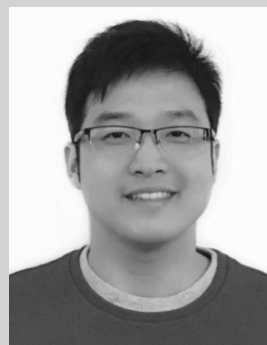
A temperature gradient is generated along the 2D material by applying a heating current I_h to the microheater through Joule heating. Then, a thermoelectric voltage V_{TEP} will be generated along this temperature gradient. The measurement of the Seebeck coefficient is thus divided into two parts: the measurement for V_{TEP} and ΔT .

1.1. ΔT Calibration

The measurement of ΔT is usually based on the four-probe resistance R of the fabricated microresistance thermometers, as R is sensitive to local temperature. To begin, these thermometers are calibrated in a cryostat with highly accurate temperature control. The resistances of each thermometer were then recorded as a function of temperature and these $R \sim T$ curves work as references in the temperature calibration process. During the ΔT measurement, a series of DC currents I_h is applied to the heater, and the corresponding resistance R of each thermometer is recorded at different heating currents. Following the relation, $R \propto \Delta T \propto I_h^2$, a parabolic $R-I_h$ curve can be obtained in this measurement. Based on the resistance differences of the thermometer and the $R \sim T$ calibration, the local temperature of each thermometer can be obtained, so that ΔT can be calculated.

1.2. V_{TEP} Measurement

There are two different methods for V_{TEP} measurement depending on whether a DC or an AC heating current is applied to the microheater. By sweeping a DC current in the microheater, a voltage difference V_{TEP} can be measured according to the heating current. A parabolic $V_{\text{TEP}}-I_h$ curve is obtained due to the relation: $V_{\text{TEP}} \propto \Delta T \propto I_h^2$. We can get a very accurate



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V_{TEP} measurement by this DC sweeping method. However, this requires a slow sweep of I_h (to ensure thermal equilibrium) and can be quite time consuming. The AC method, which typically uses a lockin amplifier to directly lock the V_{TEP} signal frequency induced by the heating current requires significantly less time. An AC heating current $I_h = I_{\text{sin}}(\omega t)$ with the

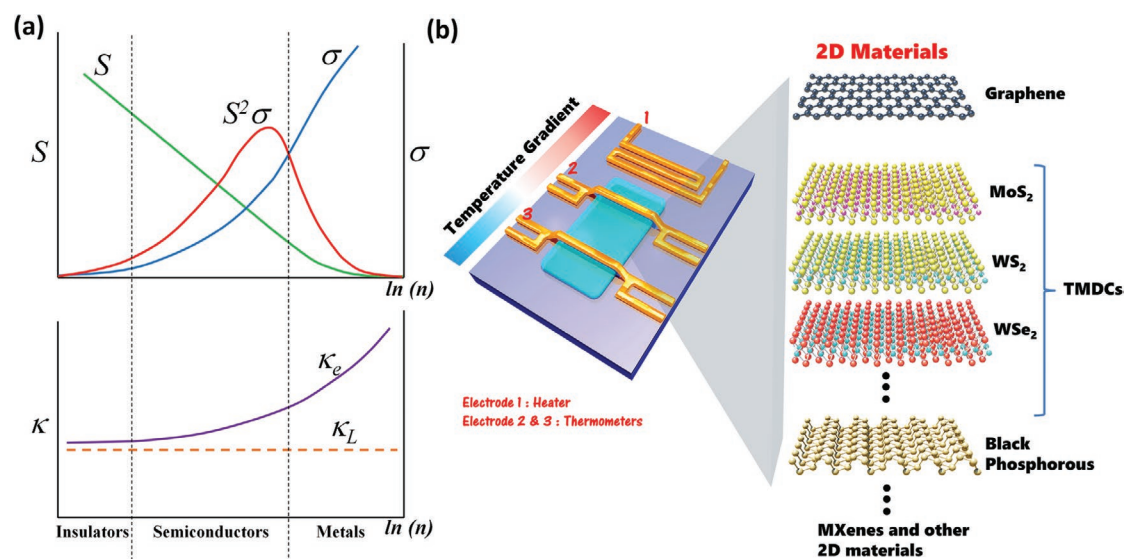


Figure 1. a) Interdependent relation of Seebeck coefficient (S), electrical conductivity (σ), and the electronic (κ_e) and lattice (κ_L) thermal conductivity for different charge carrier concentrations. Reproduced with permission.^[1] Copyright 1988, IOP Publishing. b) Illustration of the thermoelectric measurement of 2D materials.

frequency of ω is applied. As the thermoelectric voltage is proportional to the temperature difference along the 2D material $V_{\text{TEP}} \propto \Delta T \propto I_h^2 = I^2 \sin^2(\omega t)$, one can hence measure the thermoelectric voltage by sensing the 2ω signal, which is $\pi/2$ out of phase with the heater signal. However, for high resistance 2D materials, this measurement is limited by the input impedance of the lockin amplifier. After the measurement of ΔT and V_{TEP} , we can now extract the Seebeck coefficient of the measured 2D materials by $S = -V_{\text{TEP}}/\Delta T$. Employing such techniques, one can measure the in-plane thermoelectric properties of 2D materials.

Novel physics ideas can manifest in interesting thermoelectric concepts that can be tested with the 2D material platform. Thermoelectric transport measurements can therefore be used not only to predict a new generation of energy harvesting and/or thermoelectric cooling materials and devices, but also foretell new phases of matter and clarify conceptual conundrums in condensed matter physics. In this progress report, we will focus on different types of layered materials, constituted of both single-elements and compounds, and use recent experiments to suggest novel ideas and approaches toward thermoelectric physics and technology. There are already good Review articles^[15–18] that describe specifically thermal transport properties of 2D materials, and experimental measurement of thermal conductivity of suspended 2D materials is a formidable challenge that requires detailed discussions. Hence, in this Review, while we mention some recent achievements within the context of thermoelectric physics, we will focus on results and perspectives gained from measurements of the Seebeck coefficient and the thermoelectric powerfactor ($S^2\sigma$), with an eye specifically on the electronic band structure.

2. Single Element Layered Materials

2.1. Overview

Elemental materials, namely, materials composed of a single element, have relatively simple crystal and electronic structures

compared to compounds or alloys. However, a variety of allotropes could exist for many of these materials, which host a rich collection of physical properties. After breakthroughs in isolation of monolayer graphene from graphite, other elemental materials with layered structures held by vdW forces have entered the research scope. A large number of such layered elemental materials have been predicted or even experimentally demonstrated. Unlike graphene, these layered elemental materials are unstable in air, and are difficult to synthesize. Nevertheless, heavy research efforts have been devoted to them motivated by their unusual properties as well as the possibility of stabilizing them by encapsulation. **Figure 2** shows elemental materials for which layered structures have been either theoretically predicted or experimentally demonstrated. It can be seen that these elements are in the IIIA, IVA VA, or VIA group of the periodic table, and are typically nonmetals. For some of the elements, such as Te, growth of thin (≈ 30 nm), disk-like nanoplates has been demonstrated,^[19] but they are not really layered structures held by vdW interlayer forces.

Table 1 shows a summary of the bandgap of these layered elemental materials in the bulk or monolayer limit. Among the layered phase of the elemental materials, only those constituted of carbon (graphene or graphyne),^[21,22] silicon (silicene),^[23] tin (stanene),^[24] phosphorus (phosphorene),^[25] arsenic,^[26] antimony (antimonene),^[27,28] and boron (borophene)^[29] have been experimentally demonstrated, and the others still await solid experimental evidence.

Among these materials, germanene, stanene, borophene, and aluminene are predicted to be semimetals with a zero or nearly zero bandgap, and therefore coexistence of free electrons and holes are expected unless they are externally doped. Bismuthene, antimonene, arsenene, and phosphorene are predicted to have relatively wide bandgap, and hence chemical doping is needed to make them appealing for thermoelectrics. In the following, we focus on four experimentally demonstrated materials: graphene, silicene, arsenene, and phosphorene in either monolayer or multilayer forms, to discuss the current status and prospects of thermoelectric properties of layered elemental materials.

IA																	VIIIA
H	IIA											IIIA	IVA	VA	VIA	VIIA	He
Li	Be											B	C	N	O	F	Ne
Na	Mg	IIIB	IVB	VB	VIB	VIIIB	VIII	VIII	VIII	IB	IIB	Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La-Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Ac-Lr	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og

Figure 2. Yellow shaded elements are predicted or shown to have stable, layered structures.^[20]

2.2. Graphene

Graphene was the first layered material to be isolated down to the single atom level and its discovery not only led to a Nobel Prize, but also opened up a whole new library of layered materials with novel mechanical, electrical, optoelectronic, thermal properties as well as other novel physics in 2D. Unlike the carbon atoms in diamond, which are sp^3 hybridized with a bond angle of 109.5° with a tetrahedral symmetry (high bonding strength, but electrically an insulator), a graphene 2D lattice has carbon atoms that are sp^2 hybridized into a 2D plane with hexagonal symmetry (bond angle of 120°) and vdW interaction in the out-of-plane direction. This results in out-of-plane π - and π^* -bonds that form the famous linear $E-k$ dispersion relation and hence the electrons behave like massless Dirac particles with a very large Fermi velocity, $v_F \approx 10^6 \text{ m s}^{-1}$.^[21] These strong in-plane bonds give graphene immense strength^[42]

and mechanical properties,^[43] and also make its phonon thermal conductivity exceptionally high,^[44,45] as large as $3000 \text{ W m}^{-1} \text{ K}^{-1}$ for monolayer graphene at room temperature. Intuitively, this arises due to light C atoms as well as strong in-plane C–C bonds, thus reducing bond anharmonicities and resulting in large group velocities and phonon mean free paths. While this is good for thermal management and heat dissipation, it is counter-productive for thermoelectrics as it is difficult to maintain a temperature gradient in the in-plane direction, which then limits the ZT . There have been many approaches proposed both theoretically and experimentally to reduce the in-plane thermal conductivity. As one increases the number of layers of the graphene, the thermal conductivity approaches that of graphite as the out-of-plane acoustic (ZA) modes, which are unobstructed in suspended graphene, start to be scattered by the surrounding layers, hence reducing the thermal conductivity.^[46] Similarly, coupling with a SiO_2 substrate was shown to reduce

Table 1. Theoretical and experimental bandgap of layered elemental structures in the bulk or monolayer limit. The layer-dependent band structure makes the bandgap of bulk much different from the monolayer case. For the bulk, the bandgap value tends toward the limit of a large number of monolayers stacked together, unless otherwise noted as from different bulk structures.

Group	Element	Monolayer	Bulk E_g [eV]	Monolayer E_g [eV]	Exp. monolayer E_g [eV]	Refs.
IVA	C	Graphene	0	0	0	[21,30]
	Si	Silicene	1.12 tetrahedral	0.002	0.6, 0.15 (on-substrate)	[31–33]
	Ge	Germanene	0.67 tetrahedral	0.033	N/A	[31]
	Sn	Stanene	0 tetrahedral	0.1	0	[24]
VA	P	Phosphorene (black)	0.3	1.0	1.73 (on-substrate)	[34]
		Phosphorene (blue)	1.2	2	≈ 1.10 (on Au surface)	[35,36]
	As	Arsenene (gray)	0 rhombohedral	2.49	N/A	[37]
		Arsenene (black)	0.39 orthorhombic	1.54	N/A	[38]
	Sb	Antimonene	0	2.28	N/A	[37]
	Bi	Bismuthene	0	0.5	N/A	[39]
IIIA	B	Borophene	1.6–2 rhombohedral	0	0	[29]
	Al	Aluminene	0 fcc	0	N/A	[40]
VIA	Te	Tellurene	N/A	1.15 (α -Te) 1.79 (β -Te) 0 (γ -Te)	N/A	[41]

the thermal conductivity of graphene by a factor of 5, down to $600 \text{ W m}^{-1} \text{ K}^{-1}$.^[47] Owing to the large mean free paths of the phonon heat carriers, different lengths of suspended graphene were measured, resulting in ballistic (nonscattered) phonons that transmit heat from the hot to the cold side.^[48,49] Theoretically, unique to 2D graphene due to its high phonon mean free paths, liquid-like hydrodynamic flow at lower temperatures was proposed for suspended monolayers,^[50] which is different from classical Fourier's law and diffusive phonon transport. While experiments on these are challenging as one requires isotopically pure samples with large widths in order to preserve crystal momentum, we can expect that fluid-like flow of heat can provide an additional knob to tune thermal transport perhaps toward decreasing the thermal conductivity and making it favorable for thermoelectrics. In order to get a comprehensive evaluation on the thermal properties of graphene, we refer interested readers to other recent Reviews.^[15–17]

In terms of the electronic properties of graphene, naively one would expect that since the Seebeck coefficient is directly proportional to the asymmetry of density of states around the Fermi level, the magnitude would be low due to the linear dispersion relation in semimetallic graphene. The measurement techniques for in-plane Seebeck and four-probe conductivity were first demonstrated on monolayer graphene samples in 2009.^[8,10,51] Anomalous Seebeck ($S_{xx} = \Delta V_{xx}/\Delta T_{xx}$) and Nernst ($S_{xy} = \Delta V_{xy}/\Delta T_{xx}$) signals were observed, in violation of the well-accepted Mott relation for metallic samples. This was especially seen to be true close to the charge neutrality point (CNP) away from the highly doped degenerate limit, where electron-hole puddles were purported to be present (Figure 3a). In this resistive state, the carrier density is strongly inhomogeneous and also the effect of impurities was found to be nontrivial. The manifestation of this on the thermoelectric power (Seebeck and Nernst) was explained well using an effective medium theory.^[52] Predictions about the anomalously large values of Seebeck in the metallic state were also made, but the quality of the samples was not high enough to observe this violation of the Mott relation in the metallic stage (large number of carriers) because charged impurity scattering in these 2D supported films was the dominating scattering mechanism of the carriers. However, recent experiments where the graphene was processed to be extremely clean with low defect density on an atomically smooth, high- k hexagonal boron nitride (h-BN) dielectric indeed verified predictions that electron-electron scattering can dominate at high temperatures. This enabled the exploration of rich physics, where strong inelastic collisions between electrons (e-e interactions) dominate, resulting in a violation of the Mott Relation.^[53] Further, near the CNP, a plasma of electron-holes (unlike charge puddles shown previously in defect-dominated samples) manifested in large Seebeck values^[53,54] as well as a violation of the Wiedemann-Franz Law (Figure 3b).^[55] Importantly, these experimental studies showed that in such a strongly interacting regime, charge carriers can be well defined by hydrodynamics where they behave more like a liquid than a dilute gas of colliding particles. A direct experiment that observed liquid-like vortices in graphene away from the CNP showed unequivocally that this was the case in graphene even at room temperature.^[56] In general, these experiments show that thermoelectric experiments

can reveal rich physics of scattering and the consequence of electron-hole symmetry in graphene. As an example, a recent experiment on twisted bilayer graphene showed that the temperature dependence of the crossplane Seebeck coefficient reveals that the Mott relation does not hold and instead is indicative of strong electron-phonon coupling in the high-temperature limit, which could be used to probe such effects in other materials as well.^[57]

In addition to this new and interesting field of hydrodynamic electrons in graphene, to ascertain if these anomalously large Seebeck values could be favorable for thermoelectric applications, one can consider the values of the powerfactor obtained. The conductivity, σ_{2D} , increases as the gate voltage increases away from the CNP ($\approx 1 \text{ mS}$) and the dip in the Seebeck coefficient with a larger number of back gate-induced carriers is not large ($S_{\text{max}} \approx 50 \text{ } \mu\text{V K}^{-1}$). In order to compare with traditional thermoelectric (TE) materials such as bulk Bi_2Te_3 , which has a powerfactor of $\approx 3\text{--}5 \text{ mW m}^{-1} \text{ K}^{-2}$ at 300 K, we consider an equivalent 3D electrical conductivity for graphene (assuming graphene's out-of-plane thickness of $t \approx 0.35 \text{ nm}$), defined as $\sigma_{3D} \sim \sigma_{2D}/t \sim 2.9 \times 10^6 \text{ S m}^{-1}$. This yields a $\text{PF}_{3D} \approx 7 \text{ mW m}^{-1} \text{ K}^{-2}$, almost twice as large as a traditional optimally doped TE material. The bottleneck in using graphene for a TE device, however, is its extremely high thermal conductivity, therefore not allowing for a large temperature difference across the device. We note that in most thermoelectric experiments, the graphene lies on the substrate and the substrate controls the lateral temperature gradient. In recent experiments performed on high-quality graphene/h-BN, even higher values of PF_{3D} were reported, as large as $34.5 \text{ mW m}^{-1} \text{ K}^{-2}$ (converted from a $10.35 \text{ W m}^{-1} \text{ K}^{-1}$ (Figure 3c) powerfactor $\times$ temperature (PFT) value at 300 K).^[54] These large values are similar to the report of violation of the Mott Relation in clean samples and can be ascribed to hydrodynamic electrons as well.^[53] Interestingly, an active thermoelectric cooler was proposed where a material possessing a large powerfactor along with a high thermal conductivity is desirable; here graphene could indeed function as an amenable solution in applications such as local hotspot cooling in electronic devices.^[58] The maximum cooling power was shown to be proportional to PFT and in the case of graphene, for an in-plane supported thermal conductivity of $600 \text{ W m}^{-1} \text{ K}^{-1}$, this constitutes an additional $\approx 1.7\%$ of active cooling in addition to heat dissipation through the graphene layer. Complementing such an active cooler with a traditional thermoelectric refrigerator device and further optimization would constitute an interesting research direction to be explored.

Many other experiments have also been performed in graphene due to its unique capability to be tuned into both n- and p-type carriers. When light is shined on dual-gated graphene, hot carriers are generated resulting in a unique photothermoelectric effect,^[59] which could prove useful in photovoltaics and has been demonstrated to be the responsible mechanism for terahertz detection.^[60] Peltier heating (cooling) for holes (electrons) was demonstrated directly on devices, adding to the fundamental understanding of thermodynamic transport in graphene.^[61,62]

In summary, many novel concepts have been proposed and experimentally realized in graphene using thermoelectric transport as a platform. The violation of the Mott relation with an enhanced Seebeck and powerfactor in the hydrodynamic

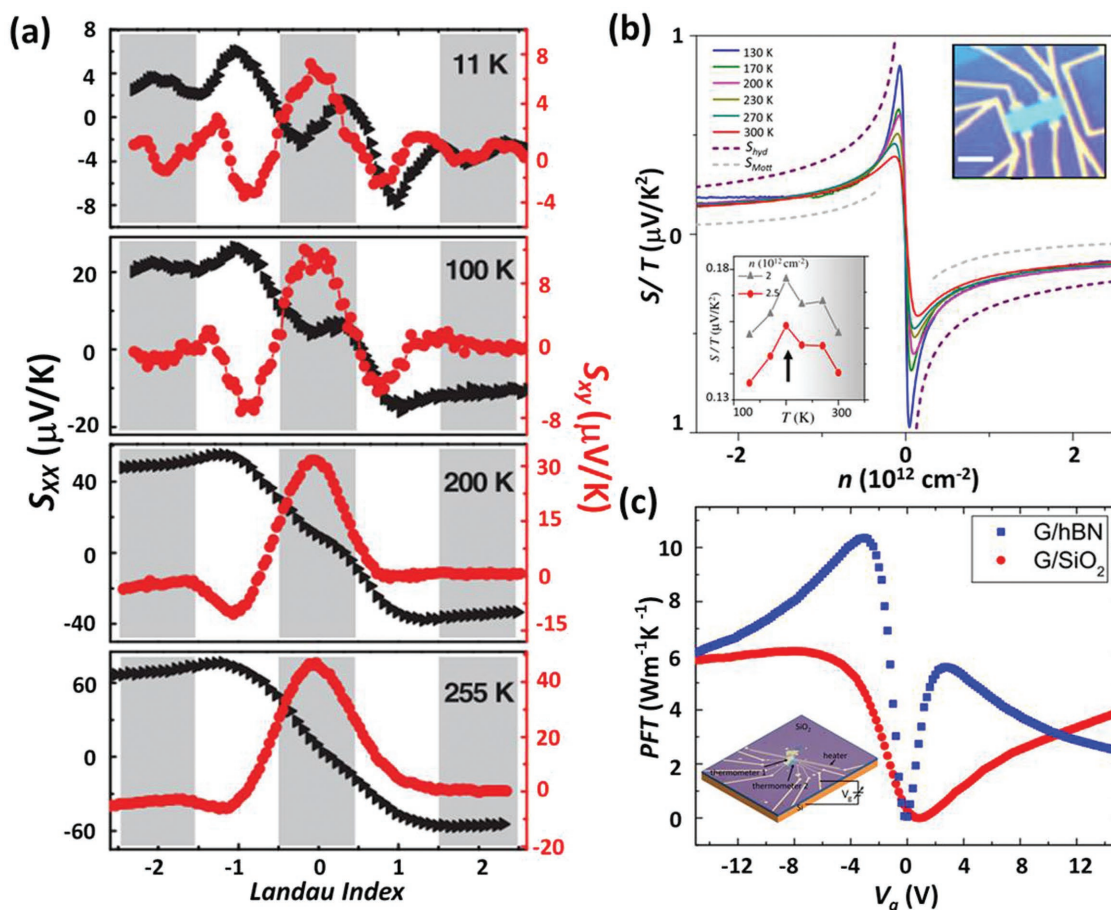


Figure 3. a) Seebeck (S_{xx}) and Nernst (S_{xy}) coefficient of graphene near the charge neutrality point at four different temperatures under magnetic field, $B = 8$ T. The Seebeck coefficient diverges with $1/\sqrt{|n_{2D}|}$ where n_{2D} is the carrier density and also the Nernst signal shows a very large peak ($\approx 50 \mu\text{V K}^{-1}$) at the Dirac point. Reproduced with permission.^[8] Copyright 2009, American Physical Society. b) Comparison between experimental data for S/T and theoretical calculation using the Mott Relation (gray dashed curves) and hydrodynamic (purple dashed curves) cases. The temperature dependent S/T is also shown in lower inset. Much larger Seebeck coefficient due to hydrodynamic transport shows alternative avenues to achieving high performance thermoelectric devices. Reproduced with permission.^[53] Copyright 2016, American Physical Society. c) Large enhancement of the powerfactor in graphene/h-BN device compared with graphene/SiO₂ device. Reproduced with permission.^[54] Copyright 2016, National Academy of Sciences.

regime could be promising for future applications, especially if engineered correctly for on-chip cooling purposes.

2.3. Phosphorene and Arsenene

Compared with zero-bandgap semimetal graphene, another monoatomic layered material, phosphorus, exists as a semiconductor. The layered forms of phosphorus have been predicted to have two allotropes, named as black^[63] and blue phosphorus,^[35,36] respectively. They differ in structure mostly in that the distinct armchair (AC) ridges in black phosphorus (Figure 4a) are converted to a zigzag (ZZ) structure in blue phosphorus.^[35] Blue phosphorus has been rarely studied, while theoretical calculations show that its monolayer has an indirect bandgap of ≈ 2 eV, and bulk has a bandgap of ≈ 1.2 eV.^[35] Black phosphorus (bP), on the other hand, is well known since at least the 1910s,^[64] and is much better understood. A renaissance of research on bP has occurred since 2014 when Li et al. reported high-performance bP field effect transistors (FETs) with a channel mobility of up

to $1000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.^[25,65] The bandgap of bP is predicted to be ≈ 0.3 eV in the bulk and ≈ 1.0 eV in the monolayer limit. Recently, the optical bandgap of bP was accurately measured to be 1.73 eV in the monolayer and 0.35 eV in the bulk.^[34]

Unlike planar graphene, the structure of bP is highly anisotropic owing to the puckered AC ridges. As a consequence, bP is shown to possess highly anisotropic physical properties. For example, the thermal conductivity (κ) along the ZZ direction of multilayer bP is about two times higher than along the AC direction at temperature higher than 100 K (Figure 4b), experimentally found to range from ≈ 10 to $\approx 25 \text{ W m}^{-1} \text{ K}^{-1}$,^[66] ≈ 10 to $\approx 50 \text{ W m}^{-1} \text{ K}^{-1}$,^[67] or from ≈ 30 to $\approx 90 \text{ W m}^{-1} \text{ K}^{-1}$.^[68] This anisotropy was attributed to the higher lattice anharmonicity and lower group velocity along the AC than the ZZ direction.^[66] Subsequent studies have shown that this anisotropy arises mainly from the dispersion relation and more recent results show that phonon relaxation time only contributes to the anisotropy in the crossplane thermal conductivity and not the in-plane (Figure 4c).^[67,69,70] However, this effect has not been clarified at low temperatures, and is yet to be studied experimentally.

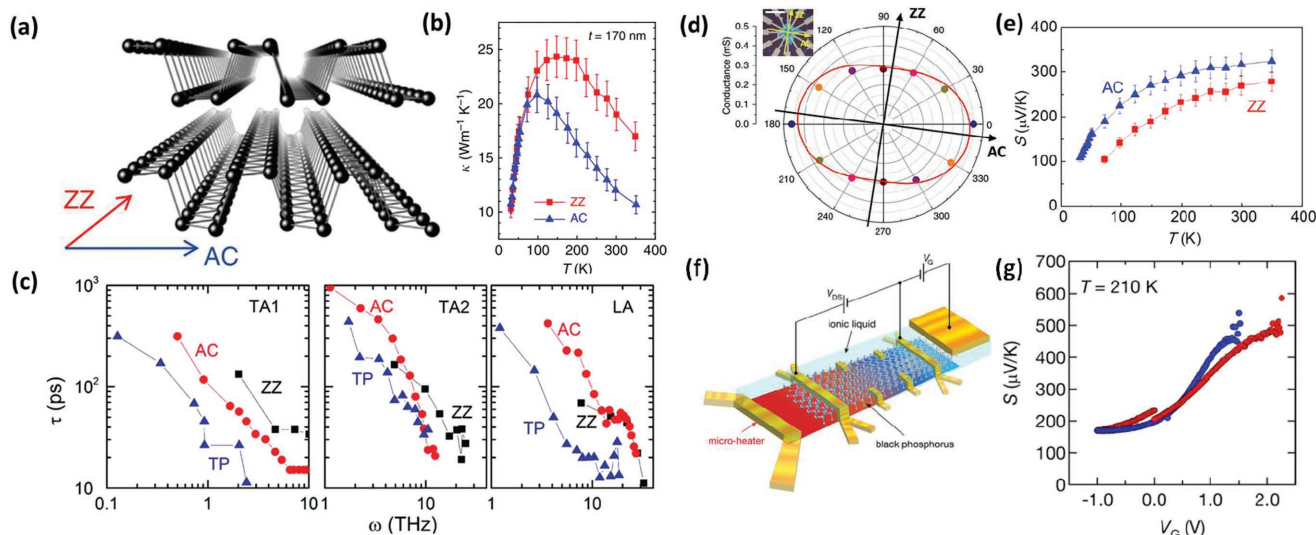


Figure 4. a) Side view of a two-layer bP showing the ZZ and AC directions. b) Anisotropic thermal conductivity of 170 nm thick bP nanoribbons. Reproduced with permission.^[66] Copyright 2015, Springer Nature. c) The relaxation times (τ) as a function of angular frequency (ω) of longitudinal (LA) and transverse (TA1 and TA2) phonon modes along the AC, ZZ, and TP (through-plane) direction of bP. Reproduced with permission.^[69] Copyright 2017, WILEY. d) Anisotropic electrical conductivity of bP. Inset is the optical image of bP flake measured here and the scale bar is 50 μm . Reproduced with permission.^[71] Copyright 2014, Springer Nature. e) Anisotropic Seebeck coefficient of the bP ribbons. Reproduced with permission.^[66] Copyright 2015, Springer Nature. These bP nanoribbons were not intentionally doped, but exhibited p-type conduction. f) Illustration of ion-gated thermoelectric measurement of bP. g) Seebeck coefficient as a function of ion-gate voltage at $T = 210$ K. Reproduced with permission.^[72] Copyright 2016, American Chemical Society.

Interestingly, the electrical conductivity (σ) of bP shows an anisotropy that is opposite to that of the thermal conductivity, namely, higher along the AC than along the ZZ direction. bP is naturally a p-type semiconductor arising from electronically active, native point defects, therefore, the anisotropy in σ is attributed to anisotropic dispersion of the top of the valence bands (Figure 4d).^[71] It has been calculated that the free hole effective mass of bP is $6.35m_0$ and $0.15m_0$ (monolayer) or $0.89m_0$ and $0.14m_0$ (five-layer) along the ZZ and AC directions, respectively,^[73] which is consistent with the higher mobility and hence conductivity in the AC direction. Seebeck coefficient (S) of bP has also been measured to be also anisotropic,^[66,74] as shown in Figure 4e. The ZT calculated from these values is very low, ≈ 0.0036 (AC) and 0.0006 (ZZ) at room temperature. This low ZT is obviously due to the relatively high κ (single-crystalline) and low σ (undoped) in the samples measured. A recent study used ion-gated techniques to control the carrier density of bP and push the S values up to $510 \mu\text{V K}^{-1}$ (Figure 4f,g).^[72] However, the ZT values still do not improve too much at this stage. On the theoretical side, bP is predicted to be a good thermoelectric material upon optimal doping^[75–78] by first-principles calculations. For example, at room temperature and optimal doping levels, bulk bP and monolayer bP (phosphorene) are predicted to have a high thermoelectric powerfactor (PF) of 11.8 and $13.9 \text{ mW m}^{-1} \text{ K}^{-2}$ respectively. A high ZT of 0.22 could be achieved in bulk, n-type bP. For phosphorene, the ZT value can reach 0.22 (n-type) and 0.30 (p-type) estimated conservatively by using the bulk thermal conductivity.^[79] Another calculation predicts $ZT \approx 1.5$ for both p- and n-type doped phosphorene with an optimal doping level of a little over 10^{16} cm^{-2} .^[75] The gap between experiments and theoretical prediction can be filled by introducing phonon

scatterers (such as intercalants, structural defects, surfaces, and nanostructuring) to bring down κ and free carriers (by chemical doping) to raise σ . Theoretical studies also suggest that uniaxial strain can significantly enhance the ZT of bP. At optimal strain levels ($\approx 7\%$), ZT values are predicted to reach 0.52 ,^[78] or 1.65 (ZZ) to 2.12 (AC) at room temperature.^[80] This enhancement is attributed to the increase in degeneracy of band valleys, or the so-called band convergence, induced by the strain. The fact that these layered and buckled materials are susceptible to high levels of strain puts them in an advantageous position over conventional 3D materials for such strain engineering in search for high thermoelectric performance.

In addition to bP, arsenic (As) also resides in group VA of the periodic table. Similar to bP, As also exists in layered structures in two different allotropes, gray Arsenic (g-As) and black arsenic (b-As). The most stable g-As is predicted to be metallic with buckled structure (space group, $P3m1$).^[81] As the cousin of bP, the puckered b-As is metastable at ambient condition (space group, $Pmna$). The calculated band gap of b-As varies significantly with the number of layers, from 0.39 eV for the bulk to 1.54 eV for the monolayer,^[38] and this is consistent with the measured absorption edge of 0.21 eV for exfoliated bulk b-As.^[26] Anisotropy in thermal conductivity of b-As was investigated using the suspended pad method. The obtained thermal conductivity along the ZZ and AC directions are 5.5 and $3.3 \text{ W m}^{-1} \text{ K}^{-1}$, respectively, at room temperature, both of which are smaller than that of b-P. The ratio, κ_{ZZ}/κ_{AC} of 1.6 , is comparable to 1.8 of b-P with similar thickness. By solving the Boltzmann transport equation, it is predicted that monolayer arsenene has higher thermal conductivity than the bulk, with 30.4 and $7.8 \text{ W m}^{-1} \text{ K}^{-1}$ along the ZZ and AC directions, respectively.^[82]

2.4. Silicene

As an isoelectronic element just like carbon, silicon can also form a layered structure. Normally, silicon takes the sp^3 tetrahedral crystal structure (diamond structure). 2D silicon, i.e., silicene, has been theoretically predicted to exist.^[83,84] Since 2010 when a number of groups reported experimental evidence of monolayer silicon stabilized on foreign substrates,^[23,32,33,85–88] research on silicene has expanded and deepened significantly. Similar to graphene, a Dirac cone band structure exists in free-standing silicene, although due to stronger spin–orbit coupling a small ($\approx 2\text{ meV}$) bandgap opens at the cone.^[31] In addition, a much wider bandgap (0.15^[33] to 0.6 eV^[32]) is seen when silicene is stabilized on a substrate. Unfortunately, currently available data on thermoelectric properties of silicene are all theoretical.^[89,90]

Owing to the nearly zero bandgap, silicene has the potential of exhibiting high electrical conductivity depending on doping, akin to that of graphene. But unlike graphene, silicene is expected to have a much lower thermal conductivity due to its heavier atomic mass and softer interatomic bonding. Using nonequilibrium Green's functions and nonequilibrium molecular dynamics simulations, it is predicted that silicene nanoribbons with edges passivated by hydrogen will have a high thermoelectric figure-of-merit, and those with AC edges usually perform better than the ones with ZZ edges.^[91] The room temperature ZT value of AC silicene nanoribbons shows a width-dependent oscillation, while it decreases slowly with increasing ribbon width for the ZZ ones. A strong temperature dependence of the thermoelectric performance was also found. The calculations show that by optimizing the doping level and temperature, their ZT values could be enhanced to as high as 4.9.^[91] In comparison, first-principles density functional techniques have been applied to silicene nanoribbons with different widths and the ZT at room temperature can reach up to 2.5.^[92] Seebeck coefficient of pure silicene at room temperature is theoretically predicted to be about $80\text{ }\mu\text{V K}^{-1}$, and not sensitive to change of temperature. By introducing a perpendicular electric field, a wide bandgap can be opened, resulting in much enhanced Seebeck coefficient.^[93]

It was also found that by controlling the type and number of edge defects, ZT values of silicene nanoribbons could be greatly enhanced.^[94] Lastly, due to the spin-polarized band structure at different valleys, valley and spin-resolved thermoelectric transport is also possible in silicene. Due to the coupling between the valley and spin degrees of freedom, pure valley and spin currents can be injected with a thermal bias.^[95] Similar effects have also been investigated with the existence of linear vacancies in silicene nanoribbons.^[96]

3. Transition Metal Dichalcogenides (TMDCs) and derivatives

3.1. Crystal and Band Structure

Following monoatomic 2D materials, a new family of layered materials, called TMDCs, has recently been explored. Their in-plane bonding is covalent, while the out-of-plane bonding

is dominated by weak vdW interactions, which allows to isolate single and few-layer samples. The first work on these materials was on optical and optoelectronic properties,^[97–101] especially studying effects such as the indirect-to-direct bandgap transition on reducing the number of layers from n to 1,^[101] spin-valley locking,^[102] valleytronics,^[103,104] and piezoelectricity,^[105,106] taking advantage of the lack of inversion symmetry in a monolayer of the TMDC. The general chemical formula for this set of compounds is MX_2 , where M is a transition metal (Mo, Hf, W, Ti, Nb, V, etc.), while X is a chalcogen (generally S, Te or Se). A plethora of rich electronic behavior has also been discovered in these 2D TMDCs including metal-insulator transitions^[107,108] and electronic correlations driving charge density waves,^[109,110] arising from not only single atomic layer studies, but also heterostructures that allow for bandgap tuning^[111] and renormalization.^[112] The metallic or semiconducting/insulating behavior is determined by how the transition metal coordination fills up the nonbonding d-states relative to the σ and σ^* bonding states, and if the Fermi level, E_F , lies inside a half-filled band or between the d-bands due to the M atoms. Generally speaking, when the metal is from group IVB and VB in the periodic table (Ti, Zr, Nb, Ta, etc.), the metal atoms obey tetragonal symmetry with one repeating unit in the c -axis direction (commonly referred to as the 1T polymorph) with octahedral coordination (Figure 5a), and E_F lies in the middle of the d-band, thus most of these 1T-TMDCs are metallic. For metals from groups V and VIB (Mo, W, etc.), the atoms obey hexagonal symmetry with two repeating units in the c -axis direction (commonly referred to as the 2H polymorph) with trigonal prismatic coordination (Figure 5b), with E_F lying in between the bands, resulting in semiconducting or insulating behavior. The 3R (rhombohedral symmetry polymorph) commonly tends to a distorted octahedral coordination of the metal atoms (Figure 5c). There are obviously exceptions to this general rule, depending on the exact nature of bonding and the size of the atoms, for example Hf (group IVB) forms an insulating 2H-HfS₂. Interestingly, the band structure can be tuned by external perturbations: for example, lithium intercalation of 2H-MoS₂ can result in metallic 1T-MoS₂.^[113]

3.2. Effect of 2D Density of States (DOS) on Thermoelectric Transport

While a large library of possible 2D semiconductors have been explored, the original hypothesis as proposed by Hicks and Dresselhaus^[114,115] has proven to be difficult to realize experimentally—a discretized electronic density of states is expected to result in an enhanced value of the thermoelectric power-factor, $S^2\sigma$, therefore enhancing the ZT . As the dimensionality decreases from 3D to 2D to 1D, the number of conduction channels reduces, thus reducing the electrical conductivity as well, but the effect of the discretized density of states enhancing the Seebeck is expected to overpower the reduced electrical conductivity.^[116–118] While electrical and thermoelectric properties of 2D semiconducting quantum wells have been studied,^[10,30,114,119,120] the electrical mobility can vary strongly as a function of the quantum well thickness, w , given by $\mu \approx \Delta E^2$, where ΔE is the

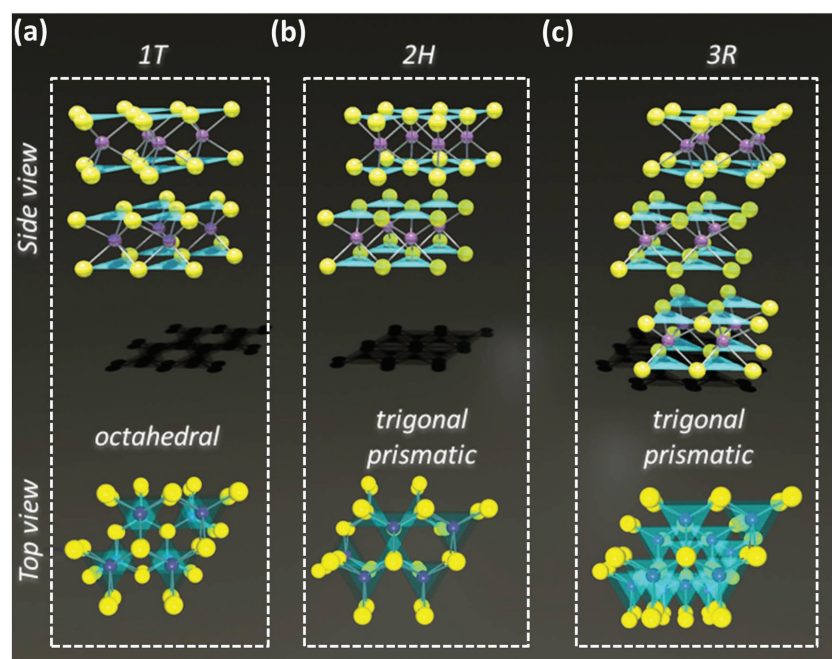


Figure 5. Top and side view of atomic arrangement and typical symmetries in a) 1T, b) 2H, and c) 3R phases of TMDCs.

perturbation in energy eigenstates of the quantum well. Here, $\Delta E \sim w^{-3}$, and hence the mobility can change drastically with the quantum well thickness, $\mu \sim w^{-6}$.^[121,122] In addition, thin films of SrTiO₃ have been studied, but oxygen vacancies and an insulating surface have restricted the mobility and the effect of confinement is challenging to observe experimentally.^[123,124]

2D TMDCs provide an attractive platform to study the effect of discretized density of states (DOS) on the thermoelectric powerfactor, mainly because of the atomically clean surfaces, lack of dangling bonds, and if prepared carefully, no roughness of the 2D surface and hence minimal quantum well thickness variation. Moreover, as high quality single and few layer TMDCs can either be synthesized^[125–127] or exfoliated^[128] on a FET device structure, different carrier concentrations can be achieved by the back gate or ionic liquid^[129–131] controlled field-effect doping, therefore potentially reducing the effect of ionized impurity scattering on the electronic and thermoelectric transport. In addition to this, the measurement of not only the on-axis Seebeck coefficient but also the transverse Nernst coefficient can be used to study important electronic effects such as phase transitions, and understanding and tuning dominant scattering mechanisms.

In order to take advantage of the 2D characteristic of the density of states to enhance the Seebeck coefficient, only semiconducting (not metallic) TMDCs are useful. To illustrate this, we look at the band structure of the semiconducting family from group VIB compounds, namely MoS₂, MoSe₂ and WS₂ and WSe₂. In the 2D limit, the transport effective mass can be written as $m_i^* = g_v \cdot g_s \cdot m_b^*$, where g_v and g_s are the valley and spin degeneracies respectively and m_b^* is the band effective mass (defined as the inverse of the 2nd derivative of energy with respect to the wavevector k in the parabolic band approximation). It is well accepted that a larger transport effective mass leads to a larger value of the Seebeck coefficient.^[7,132] In the degenerate

limit at high doping levels, we expect that σ will decrease as the effective mass increases. However, since the powerfactor is $S^2\sigma$, the favorable effect of the effective mass for S overwhelms the unfavorable effect for σ . This of course depends upon the dominant scattering mechanism, with a different energy-dependence of relaxation time for acoustic and polar optical phonons as well as charged impurities, but could be different from 3D materials, where mostly the electrons are scattering by phonons and hence the powerfactor, $S^2\sigma \approx 1/m^*$.^[7] Raamussen and Thygesen have calculated the band effective masses of both the valence and conduction bands for these 2D materials.^[133] Interestingly, exfoliated MoS₂ and MoSe₂ on SiO₂/h-BN are known to be n-type, while exfoliated WS₂ and WSe₂ are p-type, most likely arising from native defects that fix the Fermi level, E_F , close to either the conduction or the valence band.

Enhancing the density of states effective mass due to convergence of conducting bands is well-understood in many bulk materials such as doped-PbTeSe^[135] and MnSi.^[136,137] However, in 2D MoS₂, the

opportunity to tune the valley degeneracy as a function of the number of layers is unique. This is illustrated in Figure 6a for MoS₂, one of the best-studied TMDCs. Here, the transition from a bulk band structure, down to four, two and single layer MoS₂ is shown. The widely studied indirect-to-direct bandgap transition is obvious. Interestingly, as the number of layers is decreased, the conduction band minimum (CBM) and valence band maximum (VBM) are seen to shift from the $\Sigma_{\min}$ and K points, respectively, in the bulk and four-layer, both to the K-points in bilayer and monolayer. Further, while the valley degeneracy at the K-point is only 2, the valley degeneracy for the Σ direction is 6 in the conduction band, and hence the density of states effective mass resulting from a larger degeneracy at the conduction band minimum (CBM) is substantially larger.^[138] The charge density in Figure 6b also sheds some light on the layer dependence of the orbitals responsible for transport. Both at the CBM and VBM, the valleys are contributed from the molybdenum 4d states and the sulfur 2p orbitals. However, the VBM (CBM) at the K-points is significantly different from that at the $\Gamma(\Sigma_{\min})$ -points. At the K-points, the main contribution is from the S 2p_x and 2p_y in-plane orbitals and hence there is not much layer dependence (as reflected in the band structure in Figure 6a). At the Γ - and $\Sigma_{\min}$ -points, however, the DOS contribution is from the S 2p_z out-of-plane orbitals, which changes significantly as the number of layers changes due to out-of-plane coupling. For bilayer n-MoS₂, the K-point and $\Sigma_{\min}$ point are predicted to result in a net 10-fold degeneracy, therefore we expect a high value of the Seebeck from the higher DOS effective mass, and hence the powerfactor. While detailed theoretical calculations such as these are still in their infancy for other TMDC materials, the conclusion can be applied generally, where we expect a large band degeneracy to result in a large electronic powerfactor in the thin layer limit.

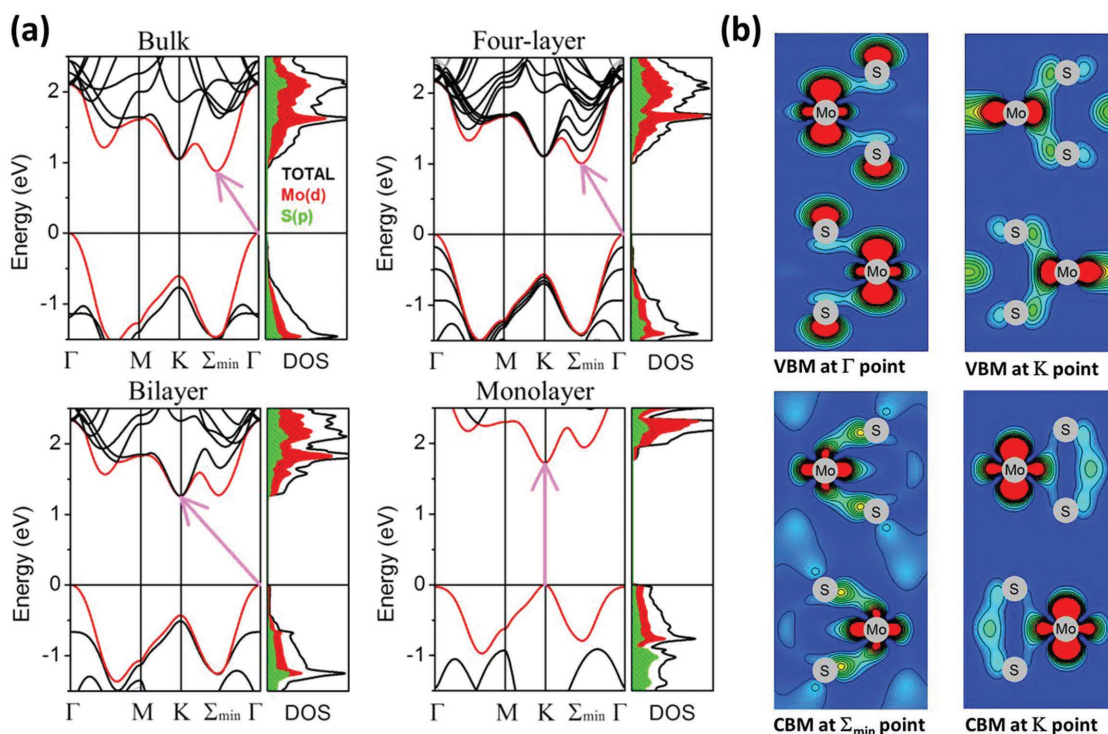


Figure 6. a) Band structures and density of states of bulk, four-layer, bilayer, and monolayer MoS₂. Only monolayer MoS₂ presents a direct band gap, of consequence for optical studies and applications, but not so important in the context of thermoelectrics. b) Charge density (red and blue represent high and low density, respectively) of bulk MoS₂ indicating the conduction/valence band minima (C/VBM). Reproduced with permission.^[134] Copyright 2016, American Physical Society.

3.3. State-of-the-Art in Thermoelectric Experiments

The experimental work on 2D crystals of TMDCs has so far centered on two compounds—MoS₂ for n-type^[108,128] and WSe₂ for p-type.^[139,140] The first measurement of the photothermoelectric effect on 1L-MoS₂ showed that a temperature gradient is produced in a monolayer MoS₂ device upon absorption of light (even at energies below the bandgap) and results in a thermovoltage (Seebeck) that drives the photocurrent through the circuit, as shown in **Figure 7a**.^[141] Here scanning photocurrent microscopy (SPCM) was performed to spatially resolve the photocurrent and the authors found an anomalously large Seebeck (as large as $-400 \mu\text{V K}^{-1}$ at a large gate voltage of 60 V in the

conducting state, and as large as -100 mV K^{-1} in the off state). The caveat is that the temperature measured at the metal/MoS₂ contacts was simulated using diffusive finite element modeling (FEM) where the interface conductance needs to be accounted for very carefully and could result in a large error. Further, it is possible that the hot electrons generating the photothermoelectric effect are not in equilibrium as the photoexcited hot carriers that diffuse in the presence of a temperature gradient do not relax with the phonons,^[11,142] therefore resulting in an anomalously small temperature (hot electrons are hotter than measured phonon temperature obtained from FEM modeling) and hence an anomalously large Seebeck coefficient. Nevertheless, the observation that the photocurrent is due to

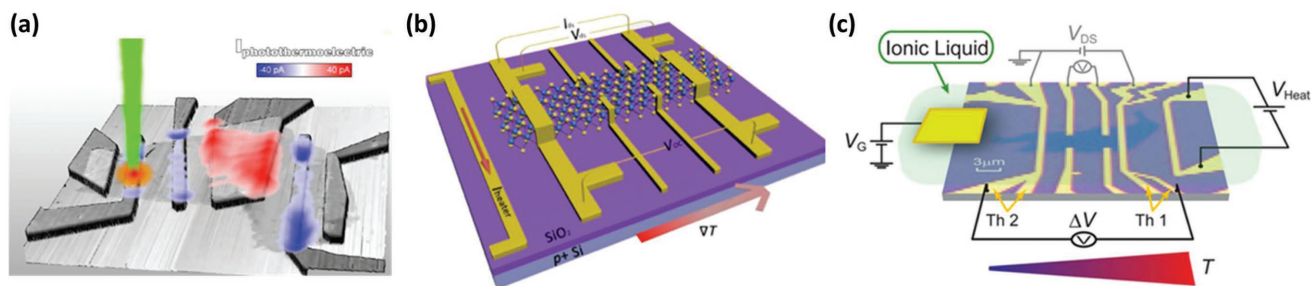


Figure 7. Illustration of a) photocurrent generation via a giant photothermoelectric effect in MoS₂ and not due to optically induced electron–hole pairs. Reproduced with permission.^[141] Copyright 2013, American Chemical Society. b) Back-gate induced thermoelectric measurement setup for monolayer, bilayer and trilayer MoS₂. Reproduced with permission.^[7] Copyright 2017, American Physical Society. c) Thermoelectric experiments on WSe₂ performed using an electric double layer (EDL) with a liquid ionic gate to tune the Seebeck with even larger carrier concentrations. Reproduced with permission.^[9] Copyright 2017, American Chemical Society.

the photothermoelectric response and not a pure photovoltaic response is interesting and warrants further studies.

The direct measurement of the thermoelectric voltage is illustrated in Figure 7b for monolayer MoS₂, the description of the technique has already been discussed in Section 1. Measurements of monolayer MoS₂ grown by the chemical vapor deposition (CVD) technique showed a Seebeck coefficient as large as 30 mV K⁻¹ in the OFF state.^[143] Further, temperature-dependent Seebeck measurements ($S \propto T^{1/3}$ in 2D) showed that transport of electrons in the MoS₂ channel followed a variable-range-hopping (VRH) mechanism, thus limiting their mobility. CVD MoS₂ has been known to have a higher density of sulfur vacancies compared to mechanically exfoliated ones, resulting in lower mobility and hence lower electrical conductivity. This is not desired for thermoelectric applications, as a high powerfactor requires not only a large Seebeck but also simultaneously high electrical conductivity. Here, metallic values of Seebeck in the ON state were reported (≈ 10 s of $\mu\text{V K}^{-1}$), which is anomalously low as it would indicate that the Fermi level is well inside the conduction band, which is unlikely even for degenerately doped MoS₂. Recent more careful work on exfoliated monolayer and few-layer MoS₂^[6,7] has revealed a high powerfactor in 2D MoS₂ (as large as 8.5 mW m⁻¹ K⁻²) and the prospect of high mobility accompanied with large Seebeck in the ON state (≈ 100 s of $\mu\text{V K}^{-1}$), stemming from high valley degeneracies, especially for bilayer MoS₂. Interestingly, the gate-voltage dependent Seebeck measurements can be used to illuminate the physics behind scattering of conduction electrons in the 2D MoS₂ channel. In both studies, energy-dependent relaxation time of the form of $\tau \propto E^r$ was used, and it was found that in monolayer MoS₂, the relaxation time is energy-independent ($r = 0$), while the two studies found a conflicting ($r = 0$ and $r = 1$) conclusion for their high-performing bilayer samples. The field-effect mobility for the higher powerfactor sample was also higher, indicating the possibility of a cleaner sample that is not affected by charged impurities, hence performing better than the lower mobility bilayer sample. However, these scattering terms can be highly sample-dependent due to the cleanliness of the samples, and more work will have to be done to isolate the exact effects of the substrate on the conduction electrons. For example, a high-k dielectric substrate with atomic scale smoothness (such as h-BN) can be used to screen the conduction electrons from the polar dielectric (typically SiO₂). This also reduces the likelihood of ripples and other impurities that could scatter the conduction electrons. A high theoretical powerfactor of 24 mW m⁻¹ K⁻² has been predicted for pristine, suspended 1L-MoS₂ with phonon-limited mobility as high as 410 cm² V⁻¹ s⁻¹ under a constant relaxation time approximation.^[144] Given that the bilayer valley degeneracy is higher, a similar calculation of bilayer MoS₂ powerfactor would probably reveal a further higher electronic powerfactor.

While the control on electron concentration of 2D MoS₂ was achieved using a back-gate configuration, a new technique illustrated in Figure 7c using an electric double layer (EDL) has recently emerged that allows to tune the carrier concentration to much higher levels ($\approx 10^{14}$ cm⁻³).^[9] Using this EDL configuration, large tuning of the Fermi level is possible.^[9] The maximum powerfactor of trilayer WSe₂ is seen to be 3.2 and 3.7 mW m⁻¹ K⁻² for n- and p-type conduction, respectively. Both Seebeck coefficient

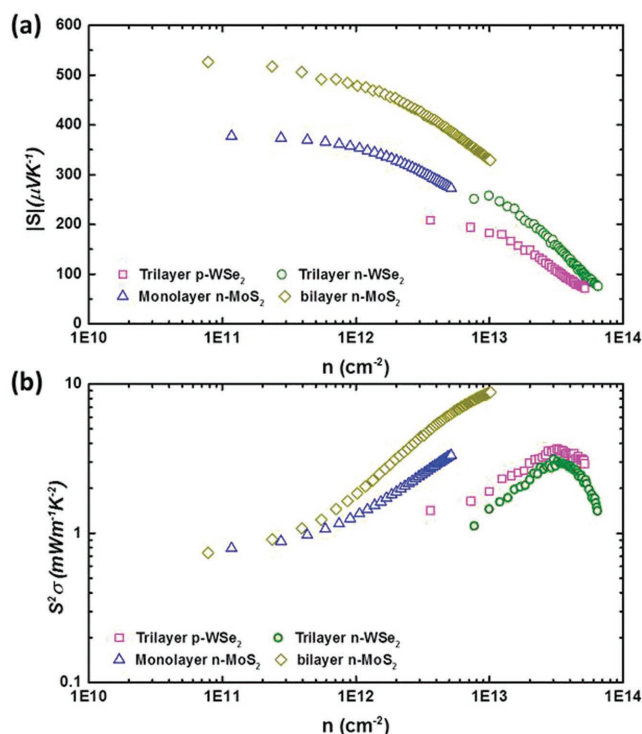


Figure 8. a) Seebeck coefficient and b) powerfactor of trilayer p-type, n-type WSe₂, and monolayer, bilayer MoS₂ as function of carrier concentration taken from data in refs.[120] and[130] respectively. While the overall trend is similar to an extrinsically doped semiconductor, the absolute values of the powerfactor scale with the mobility of the TMDC sample being measured.

and powerfactor of trilayer WSe₂ are comparable to that of n-type MoS₂ (Figure 8a,b), presumably because the effective mass and mobilities are of the same order of magnitude. Interestingly, it is also shown that the 2D powerfactor is one order of magnitude higher than the 3D powerfactor for similar levels of chemical doping, basically stemming from larger values of 2D mobilities at the same carrier concentration. This underlines the importance of 2D materials as novel material systems where the scattering can be chemically and/or electrostatically tuned, allowing for judicious control on the electronic properties, thus resulting in higher values of thermoelectric powerfactor. This also brings into question standard assumptions for high performance bulk thermoelectrics where small effective masses are favorable due to the inverse relationship with the phonon-limited mobility,^[145,146] which might not be valid in two dimensions, especially due to the ability to tune the carrier concentration as well as mobilities in new and significant ways. However, a clear, theoretical picture of how beneficial the discretized 2D DOS is for electronic powerfactor has not been presented yet, and this is surely the next step in further advancement of understanding and tuning of the thermoelectric properties.

3.4. Thermal Conductivity Engineering and Enhancement of ZT

The final piece of the puzzle lies in tuning the thermal conductivity (κ) as the thermoelectric Figure of merit is inversely

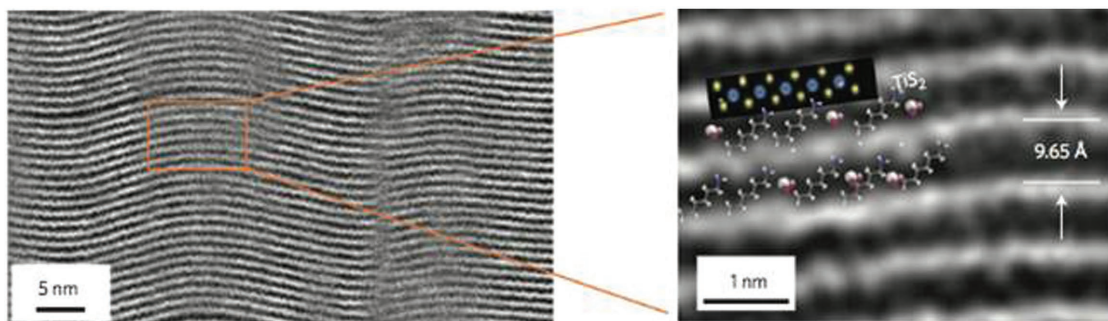


Figure 9. High-angle annular dark field scanning transmission electron microscope (HAADF-STEM) image of the $\text{TiS}_2[(\text{HA})_x(\text{H}_2\text{O})_y(\text{DMSO})_z]$ hybrid superlattice showing a wavy structure that mimics the intercalated compound. The TiS_2 is the predominant electronically conducting layer, while the low thermal conductivity due to the organic constituent enhances the ZT of the hybrid material. Reproduced with permission.^[153] Copyright 2015, Springer Nature.

proportional to the thermal conductivity, κ . κ_{total} is the sum of the electronic and phonon thermal conductivity, and at optimum doping ($n_{2D} \approx 10^{14} \text{ cm}^{-2}$, $n_{3D} \approx 10^{20} \text{ cm}^{-3}$), $\kappa_{\text{electronic}} \approx 0.1\text{--}0.4 \text{ W m}^{-1} \text{ K}^{-1}$, which is two orders of magnitude lower than the phonon thermal conductivity. For layered materials such as MoS_2 , WSe_2 , and other TMDCs, κ_{phonon} is highly anisotropic with in-plane a - and b -axis thermal conductivity much higher than the cross-plane c -axis conductivity.^[147,148] The out-of-plane ZA mode that contributes significantly to in-plane thermal conductivity in graphene^[149] is less effective in the triatom layer of TMDCs.^[150] The in-plane phonon conductivity is relatively large due to the strong covalent bonding, $\approx 36\text{--}52 \text{ W m}^{-1} \text{ K}^{-1}$ for 2D MoS_2 .^[151] Therefore, despite the large TE powerfactor, the in-plane device ZT can be quite low, $\approx 0.05\text{--}0.1$. The real challenge remains in taking advantage of the high powerfactor, while reducing the phonon conductivity using novel nanostructuring techniques.^[151,152]

Another key requirement for demonstrating functional thermoelectric devices using these novel materials is that most of the heat has to flow through the active thermoelectric element. Recently, exciting progress has been made in inorganic–organic hybrid compounds with TiS_2 (inorganic) layers electrochemically intercalated with $(\text{hexylammonium})_x(\text{H}_2\text{O})_y(\text{DMSO})_z$ organic layers, demonstrating a high electronic powerfactor of $0.45 \text{ mW m}^{-1} \text{ K}^{-2}$, shown below in **Figure 9**.^[153] While not as high as bulk TE materials (optimized $\text{Bi}_2\text{Te}_3 \approx 5 \text{ mW m}^{-1} \text{ K}^{-2}$) and other gate-doped TMDCs described earlier, the in-plane thermal conductivity of the hybrid structure was reported to be a very low value of $0.12 \text{ W m}^{-1} \text{ K}^{-1}$,^[154] resulting in an impressive ZT of 0.28 at 373 K. A large Seebeck of $-78 \text{ } \mu\text{V K}^{-1}$ was reported for the hybrid intercalation compound, which results from large DOS in the conduction channel through the TMDC TiS_2 , with three t_{2g} states of the Ti 3d orbitals. The Coulombic attractive force between hexylammonium ions and conduction electrons is weakened by the polar molecules, such as water or ethylene glycol, coexisting with hexylammonium ions which can screen the conduction electrons, leaving the mobility comparable to single crystal inorganic TiS_2 .^[155] Interestingly, the high ZT in this

hybrid material arises from the ultralow thermal conductivity, which is ascribed to stronger scattering of phonons in the TiS_2 layer due to ionic coupling to the hexylammonium ions, as compared to vdW interaction in bulk TiS_2 . This approach opens up exciting avenues for achieving a new class of TE materials: potential intercalation by changing both the inorganic TMDC layer (doped n- and p-type MoS_2 and WSe_2 for example) and organic compounds in a manner similar to this approach can result in even larger values of ZT . However, an important caveat is that in such intercalated compounds, while the overall ZT can be high, if this results from an average low thermal conductivity due to the organic part, then the amount of heat passing through the “active” thermoelectric layer, namely the TiS_2 (or similar intercalated inorganic constituent) will severely limit the efficiency of the power drawn from a thermoelectric generator. Therefore, this would require further optimization. Here we summarized few of the experimental reported powerfactors so far of different TMDCs compare to other layered 2D materials in the following **Table 2**.

Certain layered structures in their bulk form have recently been found to have extremely large ZT , such as SnSe ^[156–159] and SnS .^[160,161] their high values arise from the ultralow thermal conductivity at high temperatures due to high anharmonicity of its chemical bonds, so the discussion of those class of compounds is beyond the scope of this Review.

Finally, MXenes are obtained from the MAX phases by selective removal of the “A-element” in layered ternary $\text{M}_{n+1}\text{AX}_n$ compounds. Here M is an early transition metal and X is carbon or nitrogen.^[162,163] Theoretical predictions have shown

Table 2. Experimental reported of maximized powerfactor of layered 2D materials.

Layered 2D materials	Powerfactor [$\text{mW m}^{-1} \text{ K}^{-2}$]	Experiment conditions	Refs.
MoS_2	8.5	Bilayer MoS_2 at 300 K	[7]
WSe_2	3.7	Electric double layer increases the carrier concentration for trilayer WSe_2 at 300 K	[9]
TiS_2	0.45	intercalated with $(\text{hexylammonium})_x(\text{H}_2\text{O})_y(\text{DMSO})_z$ at room temperature	[153]
Graphene	35.7	Graphene on h-BN at 290 K	[54]
Black Phosphorous	0.46	Electric Double Layer increase the carrier concentration for few layer bP at 210 K	[72]

that due to lesser contribution from acoustic phonon branches to thermal conductivity as compared to graphene, phosphorene and TMDCs, the very low lattice thermal conductivity in functionalized MXene, Sc_2C , can enhance its ZT up to 0.3–0.5.^[164] However, experimental elucidation is required in this regard.

4. Tellurides and Selenides and the prospect of Topological Insulators (TIs)

The best single-phase materials (i.e., excluding superlattices^[165]) for near-room-temperature thermoelectrics are Bi_2Te_3 -based bulk alloys, which are also layered materials, and their best ZT is still around 1, e.g., n-type $\text{Bi}_2\text{Te}_{2.7}\text{Se}_{0.3}$ with $ZT_{\text{max}} \approx 0.9$ ^[166] and p-type $\text{Bi}_{0.5}\text{Sb}_{1.5}\text{Te}_3$ with $ZT_{\text{max}} \approx 1.2$.^[167] It is much desired to engineer these materials that would improve ZT beyond what is limited by the antagonistic coupling between the Seebeck (S) and electrical conductivity (σ). Though various experimental (e.g., energy filtering in $\text{Bi}_2\text{Te}_3/\text{Bi}_2\text{Se}_3$ superlattices^[168]) and theoretical (e.g., hybridization by topological surface states^[169]) approaches have been attempted or proposed, only S or σ , but not both, is effectively improved in these cases.

4.1. Defect-Engineering to Enhance ZT

A new way^[170,171] was discovered to drastically enhance thermoelectric properties of TIs (Bi_2Te_3 , Bi_2Se_3 , and their alloys) by utilizing native defects (NDs). This approach introduces an atomic-scale mechanism to break the trade-off between S and σ , simultaneously improving both for enhanced ZT . Such a unique combination of electrical and thermoelectric benefits originates from the multifunctionality of native point defects in Bi_2Te_3 acting as electron donors and electron energy filters. The results establish the importance of understanding and controlling point defects in thermoelectric materials as a venue to significantly improve their device performance.

The NDs were introduced by high energy particles (e.g., He^{2+} ions at $\approx 2\text{MeV}$) irradiation into polycrystalline TI thin films. Upon the irradiation, σ of Bi_2Te_3 increased for films with thickness between 47 and 740 nm, and this trend is more significant for thicker films (Figure 10). Considering the multiple conduction channels (e.g., surface and bulk) in the TI films, this effect suggests that bulk transport, which is affected by the NDs, plays an important role in the electrical conductance in this thickness range. In contrast, very thin films are insensitive to irradiation, because surface conduction dominates and remains robust to irradiation. Hall Effect measurements revealed that the enhanced σ is a combined effect of a monotonic increase in electron density (n) and a nonmonotonic change of mobility (μ). The increase in n indicated that the irradiation predominantly introduces donor-like NDs, which are also

considered as the primary reason for the unintentional n-type behavior of as-prepared Bi_2Te_3 .^[172,173] Moreover, the mobility of thick films increases remarkably (by up to 50%) upon irradiation until an intermediate dose ($\approx 2 \times 10^{14} \text{ cm}^{-2}$), then steadily decreased. The irradiation-induced NDs cause the unusual mobility behavior by modifying the relative contribution of conduction electrons between the bulk and the surface (including grain boundaries and specimen surface). Simplifying the system into a model of two electrically conduction channels, surface and bulk, the dependence of carrier concentration and mobility on irradiation dose was explained.^[174,175]

More importantly, while steadily increasing σ , the NDs at intermediate irradiation doses also improves the Seebeck coefficient (S) of the thick films as seen in Figure 10. This simultaneous enhancement of S and σ is unusual, since in most cases S decreases and σ increases with increasing n . Normally, as n increases, the Fermi level E_F moves deeper into the band where the density of states is flatter, hence reducing the entropy carried by charges around E_F .^[176] The simultaneous enhancement of S and σ is observed only in relatively thicker films ($>47 \text{ nm}$), which suggests that the measured Seebeck coefficient is dominated by the bulk contribution that can be tailored by the NDs. Using the Mott Relation, the Seebeck coefficient in

the degenerate doping limit is given by $|S| \approx \frac{k_B}{e} \frac{\pi^2}{3} \frac{k_B T}{E_F} \cdot \left(\frac{3}{2} + r \right)$,

where r is the exponent of the electron relaxation time related to kinetic energy, $\tau(\epsilon) \propto \epsilon^r$,^[177] and E_F is measured from the conduction band edge. This equation not only predicts the ordinary decrease in S as n increases (through E_F), but also an increase in S when r increases. The former leads to the conventional wisdom of the inverse coupling between S and σ , while the latter allows it to be broken, as occurs in this case. It is known that r varies from $-1/2$ for acoustic phonon scattering to $3/2$ for ionized impurity scattering in bulk.^[177] In

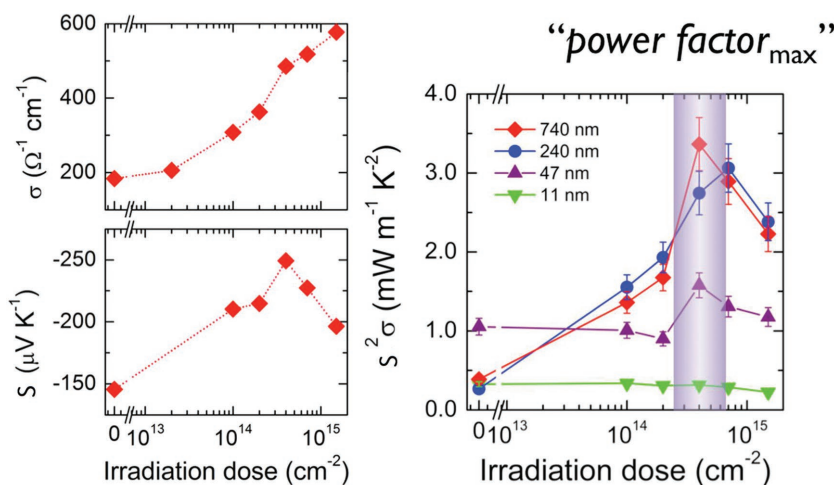


Figure 10. Simultaneous increase in electrical conductivity (up to 200%) and Seebeck coefficient (up to 70%) are demonstrated by introducing native defects in Bi_2Te_3 films, leading to a high powerfactor of $3.4 \times 10^{-3} \text{ W m}^{-1} \text{ K}^{-2}$. The maximum enhancement of powerfactor occurs when the native defects act beneficially as electron donors as well as energy filters to mobile electrons. They also act as effective phonon scatterers which reduce thermal conductivity (not shown). Reproduced with permission.^[170] Copyright 2015, Wiley.

pristine films, the measured S as a function of n follows the trend with calculation using $r = -1/2$, indicating that electrons are mostly scattered by phonons in these films. This is consistent with theoretical predictions that electrical transport in Bi_2Te_3 at similar carrier concentrations ($\approx 1 \times 10^{19} \text{ cm}^{-3}$) is limited by phonon scattering,^[178] especially considering its very large dielectric constant ($\epsilon_s = 290$).^[179] However, the high density and multiple charged states of NDs introduced by irradiation as ionized impurities cause a transition of the scattering mechanism from phonon-dominated ($r = -1/2$) toward more impurity-dominated ($r = 3/2$); as a result, the Seebeck coefficient is drastically enhanced.

The ND-enabled decoupling of S and σ naturally leads to a significant increase in the thermoelectric powerfactor, $S^2\sigma$, as shown in Figure 10. It reaches a peak value of $3.4 \pm 0.3 \text{ mW m}^{-1} \text{ K}^{-2}$ for the 740 nm film at an irradiation dose of $4 \times 10^{14} \text{ cm}^{-2}$, representing an eight-fold enhancement from its pristine value. This peak powerfactor is a factor of 1.5–3 higher compared to recently reported values in binary Bi_2Te_3 .^[168,180]

In addition, the NDs reduces the crossplane (c -axis) thermal conductivity of these films, particularly in the thick Bi_2Te_3 films, leading to a further enhancement in ZT . Therefore, irradiation-induced NDs drastically enhance thermoelectric properties in Bi_2Te_3 by decoupling the three key thermoelectric parameters and simultaneously modifying all of them toward the desired direction. The results suggest that a significant improvement of the thermoelectric performance can be achieved through judicious control of the ND species and their density by post-growth processing with high-energy beams. As the NDs are expected to be generated and behave in the similar way in a wide range of narrow-bandgap semiconductors (e.g., observed in InN ^[181] and InAs ^[182]), it is possible to extend this method to improve the figure-of-merit of other materials in conjunction with other widely utilized techniques such as alloying and nano- and heterostructuring. Although irradiation cannot be directly applied to bulk materials due to limitation in irradiation projection range ($\approx 10 \mu\text{m}$), this defect-engineering approach is of practical importance because thin film thermoelectrics could play an important role in on-chip cooling;^[183] in addition, the approach can be used in addition to existing nanoengineering techniques used in bulk thermoelectrics. For instance, nanoobjects (such as Bi_2Te_3 nanowires, particles and nanoplates) can be irradiated, and then pressed into bulk or assembled into bulk using a polymer matrix.^[184]

4.2. Surface States and Novel Thermoelectric Physics

In addition to novel nanoengineering of materials, topological phases of matter have garnered much interest in condensed matter physics. One of the first realizations of this nontrivial effect of topology was in the presence of protected 2D surface states in 3D low-bandgap layered solids, Bi_2Se_3 and Bi_2Te_3 .^[169] These surface states are metallic in nature and are protected from backscattering, with a Dirac cone crossing in the band structure, similar to graphene. Typically, this band-crossing arises from topological protection of the bulk electronic states induced by strong spin–orbit coupling resulting in no bandgap

for the surface states.^[169] However, when the top and bottom surface states hybridize (close to 4 unit cells in the c -axis direction, thickness of $\approx 1\text{--}10 \text{ nm}$), a bandgap opens up and this is predicted to manifest in enhanced ZT values of the surface states, in addition to the already high performing bulk states, especially at low temperatures;^[169] since the hybridization induced gap is $\approx 25 \text{ meV}$, this effect is dominant at temperatures T such that $k_B T \ll 25 \text{ meV}$ to prevent ambipolar transport that will decrease the Seebeck coefficient. Experimentally, the ambipolar nature of the surface states was observed in few-layer Bi_2Se_3 ^[185] by observing a sign change in the Seebeck across the CNP. Here, the Mott Relation is employed by considering the density of metallic surface states and good agreement is found with the experimental data. Most interestingly, close to the CNP, where bulk states have minimal contribution to the electrical conductivity and hence the powerfactor, a two-fold enhancement was found due to the presence of the surface states. This is despite the fact that for the 13 nm films, the hybridization gap is expected to be $\approx 10 \mu\text{eV}$ and hence the performance could be even better if thinner films were used. The interplay between novel topological phases of matter and Seebeck coefficient and thermoelectric experiments is still in nascent stages of exploration and much rich physics and innovative devices are yet to be discovered.

5. Conclusions

Many new ideas have been inspired by the novel and rich physics in layered and 2D materials, not only in fundamental thermoelectric studies but also for its practical purposes. In this progress report, we have highlighted some key achievements in thermoelectric studies of different layered 2D materials, mostly in the past decade. The violation of the Mott relation with an enhanced Seebeck and powerfactor in the hydrodynamic regime for charges in graphene shows that in addition to confinement of DOS, electron correlations and energy-dependent scattering also become exciting avenues for manipulating thermoelectric transport. Furthermore, the large effective mass induced by band convergence and degeneracy in bP and TMDCs results in high values of electronic powerfactor, suggesting an additional exploration space for high performance thermoelectric materials. Finally, the simultaneous enhancement of both Seebeck coefficient and electrical conductivity induced by native defects in Bi_2Te_3 and their alloys establishes the importance of understanding and controlling point defects in thermoelectric materials. The vast number of possibilities to optimize the thermoelectric performance in 2D layered materials will require substantial further studies in the future. For example, the creation of 2D-like bulk compounds can have 2D DOS while exhibiting low thermal conductivity and could be very exciting, especially with applications in mind. In addition, the atomically clean surfaces of layered compounds provide a novel platform to explore rich thermoelectric physics at the nanoscale: electron-phonon, photon-phonon and excitonic effects can be explored in 2D heterostructures using thermoelectric signatures. Finally, given the relatively small number of atoms in 2D, the effects of vacancies and impurities is very strong on thermoelectric transport, so correlated and magnetic

effects could also be explored. The huge library of materials as well as new physics are still being discovered, especially with the advent of high-throughput machine learning tools available today, and will surely spawn many exciting discoveries in the near future.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

2D materials, bismuth telluride, black phosphorus, graphene, layered materials, thermoelectrics, transition metal dichalcogenides

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